

1) Problem 4

2a) given: $\vec{a} = \vec{r} \ddot{\theta} - \dot{\theta}^2 \vec{r}$
 $(\vec{a} \cdot \vec{a})^2 = (\vec{a} \times \vec{a})^2 = 0$

show: $\dot{v}^2 = \dot{r}^2 + \frac{L^2}{r^2}$

$$(\vec{r} \cdot \vec{r})^2 = (\vec{r} \times \vec{r})^2 = r^4$$

$$(\vec{r} \cdot \dot{\vec{r}})^2 + \dot{\vec{r}} \cdot \dot{\vec{r}} = r^2 \dot{v}^2$$

$$(\vec{r} \cdot \dot{\vec{r}})^2 + \dot{r}^2 = r^2 \dot{v}^2$$

$$\dot{v}^2 = \dot{r}^2 + \frac{L^2}{r^2}$$

In physical terms, $\dot{r}$ = radial velocity
 $\frac{L^2}{r^2}$ = tangential velocity

$$T = 0 + h \quad \frac{1}{2} \dot{v}^2 = f(r) + h$$

$$\frac{1}{2} (r^2 + \frac{L^2}{r^2}) = f(r) + h$$

$$\frac{1}{2} (r^2 + \frac{L^2}{r^2}) = r^2 [f(r) + h]$$

$$r \dot{r}^2 + \frac{L^2}{r} = 2r^2 [f(r) + h]$$

4) show: if $0 < e < 1$ or $e > 1$

$$M a |k^2 - 1| = 0$$

$$a = \frac{r_p + r_a}{2}$$

$$r = \frac{a(1 - e^2)}{1 - e \cos f} = e \left(\frac{a}{1 - e} - r \cos f \right)$$

$$r_p = \frac{a(1 - e)}{1 - e} \quad r_a = \frac{a(1 + e)}{1 - e}$$

$$a = \frac{1}{2} \left(\frac{a(1 - e)}{1 - e} + \frac{a(1 + e)}{1 - e} \right)$$

$$2a = \frac{a(1 - e + 1 + e)}{1 - e}$$

$$2a = \frac{2a}{1 - e}$$

$$a(1 - e^2) = a$$

5a) given: $A_0 = \frac{1}{2} \pi a^2 (1 - e^2)^{3/2} \quad \dot{A} = 0$ if $e \neq 0$

show: if $0 < e < 1$ or $e > 1$, $P = \left(\frac{2\pi}{\sqrt{a}} \right) a^{3/2}$

$$A = \frac{1}{2} P$$

$$P = 2A = 2 \pi a^2 (1 - e^2)^{3/2} \quad \frac{d}{dt} \left(\frac{1}{2} \pi a^2 (1 - e^2)^{3/2} \right) = 0$$

$$= \frac{2\pi a^2 (1 - e^2)^{3/2}}{\sqrt{1 - e^2}}$$

$$= \frac{2\pi a^{3/2}}{\sqrt{1 - e^2}}$$

$$P = \frac{2\pi a^{3/2}}{\sqrt{1 - e^2}}$$

5a) define moment of inertia

$$2I = m r^2$$

$$p.d.e. \quad \ddot{I} - 2\dot{r} - \dot{u} - \dot{r} + h = u + 2h$$

$$I = \frac{1}{2} m r^2 = \frac{1}{2} m \dot{r} \cdot \dot{r}$$

$$\dot{I} = \frac{1}{2} m (\dot{r} \cdot \dot{r} + \dot{r} \cdot \dot{r})$$

$$= m \dot{r} \cdot \dot{r}$$

$$\ddot{I} = m (\dot{r} \cdot \ddot{r} + \ddot{r} \cdot \dot{r})$$

$$= m (\dot{r} \cdot \ddot{r} + \ddot{r} \cdot \dot{r}) \quad \ddot{r} = -\frac{u}{r^3} \dot{r}$$

$$= \frac{m \dot{r}^2}{2} + \dot{r} \cdot \left(-\frac{u}{r^3} \dot{r} \right)$$

$$= \frac{m \dot{r}^2}{2} + m \left(-\frac{u}{r^3} \dot{r}^2 \right)$$

$$\ddot{I} = 2\dot{r} - u = 2\dot{r} - (T + h)$$

$$\ddot{I} = 2\dot{r} + 2h - u = u + 2h$$

6a) given: u = eccentric anomaly
 f = true anomaly

$$r = a(1 - e \cos u) \quad (8.2) \quad h > 0$$

$$n(t - T) = e \sinh u - u$$

$$r = a(1 - e \cos u) \quad (8.2) \quad h < 0$$

(mean anomaly) $n(t - T) = u - e \sin u \quad (8.2) \quad h < 0$

$$r = \frac{a(1 - e^2)}{1 + e \cos f} \quad (8.2)$$

$h < 0$ (elliptical)

$$\text{trig identity: } \tan^2 \frac{u}{2} = \frac{1 - \cos u}{1 + \cos u}$$

$$\frac{a(1 - e^2)}{1 + e \cos f} = a(1 - e \cos u)$$

$$1 - e^2 = (1 - e \cos u)(1 + e \cos f)$$

$$1 - e^2 = 1 + e \cos f - e \cos u - e^2 \cos u \cos f$$

$$r = \frac{a(1 - e^2)}{1 + e \cos f} \quad \text{LHS sign}$$

$$a \cos u = a e + \frac{a(1 - e^2) \cos f}{1 + e \cos f}$$

$$a \cos u = e + \frac{a(1 - e^2) \cos f}{1 + e \cos f}$$

$$\cos u = e + \frac{a(1 - e^2) \cos f}{1 + e \cos f}$$

$$\tan^2 \frac{u}{2} = \frac{1 - \cos u}{1 + \cos u} = \frac{1 - e - \frac{a(1 - e^2) \cos f}{1 + e \cos f}}{1 + e + \frac{a(1 - e^2) \cos f}{1 + e \cos f}}$$

$$= \frac{1 + e \cos f - e - \cos f}{1 + e \cos f + e + \cos f}$$

$$= \frac{1 + e \cos f - e - \cos f}{1 + e \cos f + e + \cos f}$$

$$\begin{aligned} \text{10.1) } \frac{1}{2} \cdot \frac{1+\cos f}{1-\cos f} &= \frac{1+\cos f}{1+\cos f} \cdot \frac{1+\cos f}{1-\cos f} \\ &= \frac{1+\cos f - e - \cos f}{1+\cos f} \\ &= \frac{1+\cos f - e - \cos f}{1+\cos f + e + \cos f} \\ &= \frac{1+\cos f - e - \cos f}{1+\cos f + e + \cos f} \end{aligned}$$

$$\begin{aligned} \tan^2 f &= \frac{1-\cos f}{1+\cos f} \\ \tan^2 u &= \frac{(1-\cos f) + e(\cos f - 1)}{(1+\cos f) + e(\cos f + 1)} \\ &= \frac{(1-\cos f)(1-e)}{(1+\cos f)(1+e)} \\ &= \frac{\tan^2 \frac{f}{2}}{\tan^2 \frac{u}{2}} \end{aligned}$$

$$\tan^2 \frac{u}{2} = \frac{1-e}{1+e} \tan^2 \frac{f}{2}$$

$$\boxed{\tan \frac{u}{2} = \sqrt{\frac{1-e}{1+e}} \tan \frac{f}{2}; h < 0}$$

for $h > 0$ (hyperbolic)
use same process but for $\cosh / \tanh$

$$\boxed{\Rightarrow \tan \frac{u}{2} = \sqrt{\frac{1+e}{1-e}} \tanh \frac{f}{2}, h > 0}$$

8.4 show solutions are unique

$0 \leq e < 1$ elliptical

$$r(t-T) = u - e \sin u$$

Mean anomaly

$$f(u) = u - e \sin u$$

$$f'(u) = 1 - e \cos u \quad \cos u \in [-1, 1]$$

$$e \cos u \leq e(1) \leq 1$$

$$f'(u) = 1 - e \geq 0 \quad \forall u \Rightarrow \text{function is monotonically increasing so only 1 solution for given } t$$

$e \geq 1$

$$f(u) = e \sinh u - u$$

$$f'(u) = e \cosh u - 1 \quad \cosh(u) \geq 1 \quad \forall u \Rightarrow e \cosh u \geq e \geq 1$$

$$e \cosh u - 1 \geq 0 \quad \forall u \Rightarrow \text{function is monotonically increasing}$$

$$10.1 \quad (10.2) \quad \ell = u - e \sin u \quad 0 \leq e < 1$$

$$(10.1) \quad r = a(1 - e \cos u)$$

$$u - \ell = -e \sin u$$

periodic: $u(\ell) - \ell$
(eccentric anomaly - mean anomaly)

$$u - \ell = e \sin u$$

$$u - (\ell + 2\pi) = e \sin u$$

$$\boxed{u - \ell = e \sin u + 2\pi = e \sin u}$$

$$\text{odd: } u(-\ell) = -u(\ell) \quad -\ell = -(u - e \sin u)$$

$$-\ell = -(u - e \sin u)$$

$$\boxed{u - \ell = -e \sin u}$$

Vanishes: $\ell = 0, \ell = \pi$

$$u - 0 = e \sin u$$

$$\boxed{u = e \sin u \Rightarrow u = 0}$$

$$u - \pi = e \sin u$$

$$\boxed{u - e \sin u + \pi \Rightarrow u = \pi}$$

cont. derivative

$$\frac{d\ell}{du} = \frac{d}{du}(u - e \sin u)$$

$$1 - \frac{du}{du} - e \cos u \frac{du}{du}$$

$$d\ell = (1 - e \cos u) du$$

$$\frac{du}{d\ell} = \frac{1}{1 - e \cos u}$$

$$\frac{d(u - \ell)}{d\ell} = \frac{du}{d\ell} - 1 = \frac{1}{1 - e \cos u} - 1$$

$$\frac{d(u - \ell)}{d\ell} = \frac{1 - e \cos u}{1 - e \cos u} \rightarrow \text{continuous}$$

$$u(\ell) - \ell = \frac{e}{\pi} \int_0^\pi u_0 \sin u \, du$$

$$\text{Prove: } \frac{2}{\pi} \int_0^\pi \cos u (u - e \sin u) du$$

fourier coefficients:

$$b_n = \frac{2}{\pi} \int_0^\pi f(x) \sin(nx) dx$$

$$u_0 = \frac{2}{\pi} \int_0^\pi (u(\ell) - e) \sin(n\ell) d\ell$$

$$\text{integration by parts: } f = u(\ell) - \ell \quad g = -\frac{1}{n} \cos n\ell$$

$$f' = \frac{du}{d\ell} - 1 \quad g' = \sin n\ell$$

$$\frac{du}{d\ell} = \frac{1}{1 - e \cos u}$$

$$f' = \frac{1}{1 - e \cos u} - 1$$

$$f' = \frac{e \cos u}{1 - e \cos u}$$

$$f' d\ell = \frac{e \cos u}{1 - e \cos u} (1 - e \cos u) du$$

$$f' d\ell = e \cos u du$$

$$u_n = \frac{2}{\pi} \int_0^\pi \cos n\ell f' d\ell$$

$$= \frac{2}{\pi} \int_0^\pi \cos n\ell e \cos u du$$

$$= \frac{2}{\pi} \int_0^\pi \cos(u - e \sin u) e \cos u du$$

$$\boxed{u_n = \frac{2}{\pi} \int_0^\pi \cos(u - e \sin u) du}$$

10.2 given: q_0, q_1 pts on elliptical orbit

u_0, u_1 eccentric anomalies

$$u_1 > u_0$$

prove: distance $q_0 q_1$ is $2a \sin \frac{u_1 - u_0}{2}$

$$-\ell = -(u - e \sin u)$$

10.2 given: q_0, q_1 pts on elliptical orbit

u_0, u_1 eccentric anomalies
 $u_1 > u_0$

proj: distances $q_0 q_1$ is $2a \sin \alpha \sin \beta$

$$\alpha = \frac{1}{2}(u_1 - u_0) \quad \cos \beta = \cos \frac{1}{2}(u_1 + u_0)$$

$$\Delta \alpha \approx \Delta u \quad \beta = \frac{1}{2}(u_1 + u_0)$$

$$\vec{r} = (a \cos u - ae, a \sqrt{1-e^2} \sin u)$$

$$|\vec{r}_1 - \vec{r}_0|^2 = a^2 (\cos u_1 - \cos u_0)^2 - a^2 (1-e^2) (\sin u_1 - \sin u_0)^2$$

$$(\cos u_1 - \cos u_0)^2 = \cos^2 u_1^2 - 2 \cos u_1 \cos u_0 + \cos^2 u_0^2$$

$$(\sin u_1 - \sin u_0)^2 = \sin^2 u_1^2 - 2 \sin u_1 \sin u_0 + \sin^2 u_0^2$$

$$|\vec{r}_1 - \vec{r}_0|^2 = \cos^2 u_1 + \cos^2 u_0 - 2 \cos u_1 \cos u_0 - (1-e^2) (\sin^2 u_1 + \sin^2 u_0 - 2 \sin u_1 \sin u_0)$$

$$\cos u_1 - \cos u_0 = -2 \sin \frac{u_1 + u_0}{2} \sin \frac{u_1 - u_0}{2}$$

$$\sin u_1 - \sin u_0 = 2 \cos \frac{u_1 + u_0}{2} \sin \frac{u_1 - u_0}{2}$$

$$|\vec{r}_1 - \vec{r}_0|^2 = 4 \sin^2 \alpha \sin^2 \beta + 4(1-e^2) \cos^2 \beta \sin^2 \alpha$$

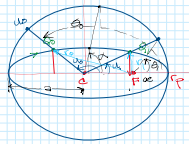
$$= 4 \sin^2 \alpha (\sin^2 \beta + (1-e^2) \cos^2 \beta)$$

$$= 4 \sin^2 \alpha (\sin^2 \beta + \cos^2 \beta - e^2 \cos^2 \beta)$$

$$= 4 \sin^2 \alpha (1 - e^2 \cos^2 \beta)$$

$$|\vec{r}_1 - \vec{r}_0|^2 = 4 \sin^2 \alpha \sin^2 \beta \cos^2 \beta$$

$$q_0 q_1 = 2 \sin \alpha \sin \beta$$



10.2 $\vec{r} = -\frac{\mu}{r^2} \hat{r}$ (14.2)

$$(\vec{r} \cdot \vec{r})' = 2(\mu/r^2) \quad \dot{r} = 0 \quad \mu = G(m_1 m_2)$$

$$(\vec{r} \cdot \vec{r})' = 2(\mu/r^2) \quad r \rightarrow 0 \quad \vec{r} \rightarrow \vec{r}_1$$

$$\text{Show } |\dot{\vec{r}}|^2 \rightarrow \frac{2\mu}{r} \quad r(t \rightarrow t_1) \rightarrow (\frac{1}{2}\mu)^{1/2} \text{ as } t \rightarrow t_1$$

$$r \dot{r}^2 = 2\mu + 2\mu \dot{r}^2$$

$$\sqrt{r \dot{r}^2} \rightarrow \sqrt{2\mu}$$

$$|\dot{\vec{r}}|^2 \rightarrow \sqrt{2\mu} \quad \checkmark$$

$$\frac{d\vec{r}}{dt} = \frac{d\vec{r}}{dt}$$

$$\int r^{1/2} dr = \int \sqrt{2\mu} dt$$

$$\frac{2}{3} r^{3/2} = \sqrt{2\mu} t + C$$

$$\frac{2}{3} r^{3/2} = \sqrt{2\mu} (t - t_1)$$

$$r^{3/2} = \frac{3}{2} \sqrt{2\mu} (t - t_1)$$

$$r = \left[\frac{3}{2} \sqrt{2\mu} (t - t_1) \right]^{2/3}$$

$$r = \left(\frac{3}{2} \mu \right)^{1/3} (t - t_1)^{2/3}$$

$$\frac{r}{(t - t_1)^{2/3}} = \left(\frac{3}{2} \mu \right)^{1/3}$$

$$r(t - t_1)^{-2/3} \rightarrow \left(\frac{3}{2} \mu \right)^{1/3} \text{ as } t \rightarrow t_1$$

10.1 $\vec{F} = \frac{\mu}{r^2} \hat{r} + \frac{\mu}{r^2} \frac{d\vec{r}}{dt}$ (14.2) $\vec{r} \times (-\frac{\mu}{r^2} \hat{r}) = \vec{r} \times \vec{F}$

$$\vec{r} \times \left(\frac{\mu}{r^2} \hat{r} + \frac{\mu}{r^2} \frac{d\vec{r}}{dt} \right) = \frac{\mu}{r^2} \vec{r} \times \frac{d\vec{r}}{dt} + \frac{\mu}{r^2} \vec{r} \times \vec{r} = \frac{\mu}{r^2} \vec{r} \times \frac{d\vec{r}}{dt}$$

$$\vec{r} \times \frac{d\vec{r}}{dt} = \frac{d}{dt} \left(\frac{1}{2} \vec{r} \times \vec{r} \right) = \frac{d}{dt} \left(\frac{1}{2} \vec{r} \times \vec{r} \right) = \frac{d}{dt} \left(\frac{1}{2} \vec{r} \times \vec{r} \right)$$

$$\vec{r} \times \frac{d\vec{r}}{dt} = \frac{d}{dt} \left(\frac{1}{2} \vec{r} \times \vec{r} \right) = \frac{d}{dt} \left(\frac{1}{2} \vec{r} \times \vec{r} \right) = \frac{d}{dt} \left(\frac{1}{2} \vec{r} \times \vec{r} \right)$$

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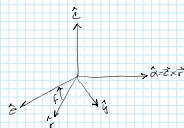
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Hwk 1 Prob 2

Friday, January 23, 2026 10:20 PM

2) Linear momentum $\vec{p} = \sum_{i=1}^N m_i \vec{v}_i$

angular momentum $\vec{H} = \sum_{i=1}^N m_i \vec{r}_i \times \vec{v}_i$

System Energy $E = \frac{1}{2} \sum_{i=1}^N m_i v_i^2 + U$

$$\dot{\vec{p}} = \sum_{i=1}^N m_i \dot{\vec{v}} = \sum m_i \vec{a}$$

Linear momentum:

$$F_{ij} = \frac{G m_i m_j}{r_{ij}^3} \vec{r}_{ji} = - \frac{G m_i m_j}{r_{ij}^3} \vec{r}_{ij}$$

$$\sum_{i=1}^N \vec{F}_i = \sum_{i=1}^N \sum_{j=1, j \neq i}^N \frac{G m_i m_j}{|\vec{r}_{ij}|^3} \vec{r}_{ji}$$

$$= \sum \left(\frac{m_i m_j}{r_{ij}^3} \vec{r}_{ji} + \frac{m_i m_j}{r_{ij}^3} \vec{r}_{ij} \right)$$

$$= \sum \frac{m_i m_j}{r_{ij}^3} (\vec{r}_{ji} + \vec{r}_{ij}) = 0$$

angular momentum

$$\vec{H} = \sum_{i=1}^N m_i \vec{r}_i \times \vec{v}_i$$

$$\dot{\vec{H}} = \sum m_i (\dot{\vec{r}}_i \times \vec{v}_i + \vec{r}_i \times \dot{\vec{v}}_i)$$

$$\dot{\vec{H}} = \sum_{i=1}^N m_i (\vec{r}_i \times \vec{a}_i)$$

$$= \sum_{i=1}^N \sum_{j=1, j \neq i}^N \vec{r}_i \times \vec{F}_{ij}$$

$$= \sum_{i=1}^N \sum_{j=1, j \neq i}^N \vec{r}_i \times \left(- \frac{G m_i m_j}{|\vec{r}_{ij}|^3} \vec{r}_{ij} \right)$$

$$= \sum_{i=1}^N \sum_{j \neq i}^N \vec{F}_{ij} \times \left(-\frac{1}{|\vec{r}_{ij}|^3} \vec{r}_{ij} \right)$$

$$\vec{r}_{ij} \times \vec{r}_{ij} = 0$$

$$\vec{H} = 0$$

Energy

$$\sum_{i=1}^N \sum_{\substack{j=1 \\ i \neq j}}^N \frac{m_i m_j}{r_{ij}} = 2 \sum_{i < j} \frac{m_i m_j}{r_{ij}}$$

$$U = -\frac{G}{2} \cdot 2 \sum_{i < j} \frac{m_i m_j}{r_{ij}} = -G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}}$$

total energy

$$T = \frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i$$

$$\frac{dT}{dt} = \frac{1}{2} \sum_{i=1}^N m_i (\dot{\vec{r}}_i \cdot \ddot{\vec{r}}_i + \ddot{\vec{r}}_i \cdot \dot{\vec{r}}_i)$$

$$\frac{dT}{dt} = \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \ddot{\vec{r}}_i$$

$$\vec{F}_i = m_i \ddot{\vec{r}}_i = \sum_{j=1, j \neq i}^N \frac{G m_i m_j}{r_{ij}^3} \vec{r}_{ji}$$

$$\frac{dT}{dt} = \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{G m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ji}$$

$$\frac{dU}{dt} = -\frac{G}{2} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{d}{dt} \left(\frac{m_i m_j}{\|\vec{r}_{ij}\|} \right)$$

$$= -\frac{G}{2} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N m_i m_j \left(\frac{1}{\|\vec{r}_{ij}\|^2} \right) \frac{d \|\vec{r}_{ij}\|}{dt} \rightarrow \sqrt{\vec{r}_{ij} \cdot \vec{r}_{ij}}$$

$$= +\frac{G}{2} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N m_i m_j \frac{1}{\|\vec{r}_{ij}\|^2} \left(\frac{1}{2} \sqrt{\vec{r}_{ij} \cdot \vec{r}_{ij}} \right) \frac{d(\vec{r}_{ij} \cdot \vec{r}_{ij})}{dt}$$

$$= -\frac{G}{2} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} (\dot{\vec{r}}_i - \dot{\vec{r}}_j) \cdot \dot{\vec{r}}_{ij}$$

$$\frac{dU}{dt} = -G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \dot{\vec{r}}_{ij}$$

$$\begin{aligned} \frac{d(T+U)}{dt} &= \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{G m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \dot{\vec{r}}_{ji} \\ &\quad - \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{G m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \dot{\vec{r}}_{ji} = 0 \end{aligned}$$

$$3) U = -G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\vec{r}_{ij}\|}$$

$$U = -G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\alpha \vec{r}_{ij}\|} = -\frac{G}{\alpha} \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\vec{r}_{ij}\|} \Rightarrow n = -1$$

$$\frac{\partial U}{\partial \vec{r}_i} = -G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}^3} \vec{r}_i \cdot \vec{r}_{ij}$$

$$\vec{r}_i \cdot \frac{\partial U}{\partial \vec{r}_i} = G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}^3} \underbrace{\vec{r}_i \cdot \vec{r}_i}_{r_i^2} \cdot \vec{r}_{ij}$$

$$\vec{r}_i \cdot \frac{\partial U}{\partial \vec{r}_i} = G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}} \vec{r}_{ij} = -U$$

$$T = \frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i$$

$$T = \frac{1}{2} \sum_{i=1}^N m_i \alpha \dot{\vec{r}}_i \cdot \alpha \dot{\vec{r}}_i = \alpha^2 \left(\frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i \right) \Rightarrow n = 2$$

$$\frac{\partial T}{\partial \vec{r}_i} = \sum_{i=1}^N m_i \dot{\vec{r}}_i$$

$$\vec{r}_i \cdot \frac{\partial T}{\partial \vec{r}_i} = \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i = 2T$$

$$I_p = \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i$$

$$I_p = \sum_{i=1}^N m_i \alpha \vec{r}_i \cdot \alpha \vec{r}_i = \alpha^2 r_i \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i \Rightarrow N = 2$$

$$\frac{\partial I_p}{\partial \vec{r}_i} = 2 \sum_{i=1}^N m_i \vec{r}_i$$

$$\vec{r}_i \cdot \frac{\partial I_p}{\partial \vec{r}_i} = 2 \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i = 2I_p$$

$$I_p U^2 = \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i \left[G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}} \right]^2$$

$$= \sum_{i=1}^N m_i \alpha \vec{r}_i \cdot \alpha \vec{r}_i \left(G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\alpha r_{ij}} \right)^2 \Rightarrow n = 0$$

for homogeneous functions order product of functions equals sum of order $(2 + 2(-1) = 0)$
so order = 0

$$4) \vec{H} = \sum_{i=1}^N m_i \vec{r}_i \times \dot{\vec{r}}_i$$

$$H^2 \leq 2T I_P$$

$$a = \vec{r} \quad b = \dot{\vec{r}}$$

$$\vec{H} = \sum m_i a \times b$$

$$\vec{H}^2 = \sum m_i a \times b \sum m_i a \times b$$

$$= \sum m_i^2 (a \times b)(a \times b)$$

$$H^2 = \sum m_i^2 [a^2 b^2 - (a \times b)(b \times a)]$$

$$H^2 \leq 2T I_P$$

$$T = \frac{1}{2} \sum m_i a^2$$

$$I_P = \sum m_i b^2$$

$$A = 2T$$

$$B = I_P$$

Hwk 1 Prob 5

Friday, January 23, 2026

10:22 PM

5)

Lagrange Identity

$$\left(\sum_{k=1}^N a_k^2\right) \left(\sum_{k=1}^N b_k^2\right) - \left(\sum_{k=1}^N a_k b_k\right)^2 = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N (a_i b_j - a_j b_i)^2$$

$$I_P = \sum_{i=1}^N m_i r_i^2 = \sum_{i=1}^N m_i (\vec{r}_i \cdot \vec{r}_i)$$

$$\vec{r}_{ij} = \vec{r}_i - \vec{r}_j$$

$$\frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (\vec{r}_i - \vec{r}_j)^2$$

$$\frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (r_i^2 - 2\vec{r}_i \cdot \vec{r}_j + r_j^2)$$

$$\frac{1}{2M} \left(\sum_{i=1}^N m_i r_i^2 \sum_{j=1}^N m_j - 2 \sum_{i=1}^N \sum_{j=1}^N \vec{r}_i \cdot \vec{r}_j + \sum_{j=1}^N m_j \sum_{i=1}^N m_i r_j^2 \right)$$

$$\underbrace{\frac{1}{2} \sum_{i=1}^N m_i r_i^2}_{I_P} - \underbrace{\frac{1}{M} \sum_{i=1}^N m_i \vec{r}_i \cdot \sum_{j=1}^N m_j \vec{r}_j}_{M \vec{R}_c \cdot \vec{R}_c} + \underbrace{\frac{1}{2M} \sum_{i=1}^N m_i \sum_{j=1}^N m_j r_j^2}_{I_P}$$

$$\frac{1}{2} I_P + \frac{1}{2} I_P - M \vec{R}_c \cdot \vec{R}_c$$

$$\frac{1}{2M} \sum_{i,j} m_i m_j \vec{r}_i \cdot \vec{r}_j = I_P - M \vec{R}_c \cdot \vec{R}_c$$

$$I_P = \frac{1}{2M} \sum_{i,j} m_i m_j \vec{r}_i \cdot \vec{r}_j + M \vec{R}_c \cdot \vec{R}_c$$

$$T = \frac{1}{2} \sum_{i=1}^N m_i v_i^2$$

$$\frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j v_{ij}^2 = \frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (v_i - v_j)^2$$

$$= \frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (v_i^2 - 2\vec{v}_i \cdot \vec{v}_j + v_j^2)$$

$$= \frac{1}{2M} \left(\sum_{i=1}^N \sum_{j=1}^N m_i m_j v_i^2 - \sum_{i,j} 2m_i m_j \vec{v}_i \cdot \vec{v}_j + \sum_{i,j} m_i m_j v_j^2 \right)$$

$$= \frac{1}{2M} \left(\underbrace{\sum_{i=1}^N m_i v_i^2 \sum_{j=1}^N m_j}_{I_P} - \sum_{i=1}^N m_i \vec{v}_i \cdot \sum_{j=1}^N m_j \vec{v}_j + \underbrace{\sum_{i=1}^N m_i \sum_{j=1}^N m_j v_j^2}_{I_P} \right)$$

$$= \frac{1}{2M} \left(\underbrace{\sum_{i=1}^N m_i v_i^2}_T \underbrace{\sum_{j=1}^N m_j}_M \underbrace{\sum_{i=1}^N m_i \vec{v}_i \cdot \sum_{j=1}^N m_j \vec{v}_j}_{\vec{v}_c} + \underbrace{\sum_{i=1}^N m_i}_M \underbrace{\sum_{j=1}^N m_j v_j^2}_T \right)$$

$$= \frac{1}{2} (2T - M \vec{v}_c \cdot \vec{v}_c)$$

$$T = \frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j v_{ij}^2 + \frac{1}{2} M \vec{v}_c \cdot \vec{v}_c$$

6) show:

$$\begin{aligned}
 \sum_{i=1}^N m_i \tilde{\mathbf{r}}_i \cdot \tilde{\mathbf{v}}_i &= \frac{1}{M} \sum_{i=1}^N \sum_{j=i+1}^N m_i m_j \tilde{\mathbf{r}}_{ij} \cdot \tilde{\mathbf{v}}_{ij} \\
 &= \frac{1}{M} \sum_{i=1}^N \sum_{j=i+1}^N m_i m_j (\tilde{\mathbf{r}}_i - \tilde{\mathbf{r}}_j) \cdot (\tilde{\mathbf{v}}_i - \tilde{\mathbf{v}}_j) \\
 &= \frac{1}{M} \sum_{i=1}^N \sum_{j=i+1}^N m_i m_j [\tilde{\mathbf{r}}_i \cdot \tilde{\mathbf{v}}_i + \tilde{\mathbf{r}}_j \cdot \tilde{\mathbf{v}}_j - \tilde{\mathbf{r}}_j \cdot \tilde{\mathbf{v}}_i - \tilde{\mathbf{r}}_i \cdot \tilde{\mathbf{v}}_j] \\
 &= \frac{1}{M} \left[\sum_{i=1}^N m_i \tilde{\mathbf{r}}_i \sum_{j=i+1}^N m_j + \sum_{j=i+1}^N m_j \tilde{\mathbf{r}}_j \sum_{i=1}^N m_i - \sum_{i=1}^N m_i \tilde{\mathbf{v}}_i \sum_{j=i+1}^N m_j \tilde{\mathbf{r}}_j - \sum_{j=i+1}^N m_j \tilde{\mathbf{v}}_j \sum_{i=1}^N m_i \tilde{\mathbf{r}}_i \right] \quad R_i = \sum m_i \mathbf{r}_i \\
 &= \frac{1}{M} \left[M \sum_{i=1}^N m_i \tilde{\mathbf{r}}_i \cdot \tilde{\mathbf{v}}_i - M \sum_{j=1}^N m_j \tilde{\mathbf{r}}_j \cdot \tilde{\mathbf{v}}_j + \sum_{i=1}^N \sum_{j=i+1}^N m_i m_j \tilde{\mathbf{r}}_i \cdot \tilde{\mathbf{v}}_j + \sum_{j=1}^N \sum_{i=j+1}^N m_i m_j \tilde{\mathbf{r}}_j \cdot \tilde{\mathbf{v}}_i \right]
 \end{aligned}$$

Hwk 1 Prob 7

Friday, January 23, 2026 10:23 PM

$$7) M_1 = m_1 \quad M_2 = m_1 + m_2 \quad M_3 = m_1 + m_2 + m_3$$

$$\vec{R}_1 = \vec{r}_2 - \vec{r}_1 \quad \vec{r} = \vec{R}_2 = \vec{r}_3 - \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \quad R_{cm} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3}{M}$$

$$T = \frac{1}{2} \sum_{i=1}^3 m_i \dot{\vec{r}}_i^2$$

define $M_i = \sum_{j=1}^N m_j$; $\vec{R}_N \equiv$ system center of mass

$$T = \frac{1}{2} \sum_{i=1}^{N-1} M_{i-1} \frac{M_{i-1} m_i}{M_i} \dot{\vec{R}}_i \cdot \dot{\vec{R}}_i + \frac{1}{2} M_N \dot{\vec{R}}_N \cdot \dot{\vec{R}}_N$$

$$T = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} \dot{\vec{R}} \cdot \dot{\vec{R}} + \frac{1}{2} \frac{m_3 (m_1 + m_2)}{m_1 + m_2 + m_3} \dot{\vec{r}} \cdot \dot{\vec{r}} \quad \text{KE of COM}$$

$$\mu_1 = \frac{m_1 m_2}{M_2} \quad \mu_2 = \frac{M_2 m_3}{M_3}$$

$$\text{COM: } \vec{R}_1 = \vec{r}_2 - \vec{R}_1^c = \vec{r}_2 - \vec{r}_1$$

$$\vec{R}_2 = \vec{r}_3 - \vec{R}_2^c = \vec{r}_3 - \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

$$\vec{r}_3 = \vec{R}_2^c + \vec{r}$$

$$\vec{r}_2 = \vec{R}_2^c + \frac{m_1}{M_2} \vec{R}_1$$

$$\vec{r}_1 = \vec{R}_2^c - \frac{m_2}{M_2} \vec{R}_1$$

$$U = - \frac{G m_1 m_2}{|\vec{R}|} - \frac{G m_1 m_3}{|\vec{r} + \frac{m_2}{M_2} \vec{R}|} - \frac{G m_2 m_3}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|}$$

$$L = T - U$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\vec{R}}} = - \frac{\partial U}{\partial \vec{R}}$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\vec{R}}} = \frac{m_1 m_2}{M_2} \ddot{\vec{R}}$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\vec{r}}} = \frac{m_3 M_2}{M_3} \ddot{\vec{r}}$$

$$- \frac{\partial}{\partial \vec{R}} \left(- \frac{G m_1 m_2}{R} \right) = - G m_1 m_2 \frac{\vec{R}}{R^3}$$

$$- \frac{\partial}{\partial \vec{R}} \left(- \frac{G m_1 m_3}{\vec{r} + \frac{m_2}{M_2} \vec{R}} \right) = - G m_1 m_3 \frac{m_2}{M_2} \frac{(\vec{r} + \frac{m_2}{M_2} \vec{R})}{|\vec{r} + \frac{m_2}{M_2} \vec{R}|^3}$$

$$- \frac{\partial}{\partial \vec{R}} \left(- \frac{G m_2 m_3}{\vec{r} - \frac{m_1}{M_2} \vec{R}} \right) = - G m_2 m_3 \frac{(-m_1)}{M_2} \frac{(\vec{r} - \frac{m_1}{M_2} \vec{R})}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|^3} = \frac{G m_1 m_2 m_3}{M_2} \frac{(\vec{r} - \frac{m_1}{M_2} \vec{R})}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|^3}$$

$$- \frac{\partial U}{\partial \vec{R}} = \left[- \frac{G m_1 m_2}{R^3} \vec{R} - \frac{G m_1 m_2 m_3}{M_2} \frac{(\vec{r} + \frac{m_2}{M_2} \vec{R})}{|\vec{r} + \frac{m_2}{M_2} \vec{R}|^3} - \frac{G m_2 m_3}{M_2} \frac{(\vec{r} - \frac{m_1}{M_2} \vec{R})}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|^3} \right] = \frac{m_1 m_2}{M_2} \ddot{\vec{R}}$$

$$-\frac{\partial U}{\partial \vec{R}} = \left[-\frac{Gm_1 m_2}{R^3} \vec{R} - \frac{Gm_1 m_2 m_3}{M_2} \left[\frac{\vec{S} + \frac{m_3}{M_2} \vec{R}}{|\vec{S} + \frac{m_3}{M_2} \vec{R}|^3} - \frac{\vec{S} - \frac{m_1}{M_2} \vec{R}}{|\vec{S} - \frac{m_1}{M_2} \vec{R}|^3} \right] \right] = \frac{m_1 m_2}{M_2} \vec{R}$$

$$-\frac{\partial U}{\partial \vec{S}} \left(-\frac{Gm_1 m_3}{|\vec{S} + \frac{m_3}{M_2} \vec{R}|^3} \right) = -Gm_1 m_3 \frac{\vec{S} + \frac{m_3}{M_2} \vec{R}}{|\vec{S} + \frac{m_3}{M_2} \vec{R}|^3}$$

$$-\frac{\partial U}{\partial \vec{S}} \left(-\frac{Gm_2 m_3}{|\vec{S} - \frac{m_1}{M_2} \vec{R}|^3} \right) = -Gm_2 m_3 \frac{\vec{S} - \frac{m_1}{M_2} \vec{R}}{|\vec{S} - \frac{m_1}{M_2} \vec{R}|^3}$$

$$\frac{\partial U}{\partial \vec{S}} = \frac{m_3 M_2}{M_3} \ddot{\vec{S}} = -Gm_1 m_3 \frac{\vec{S} + \frac{m_3}{M_2} \vec{R}}{|\vec{S} + \frac{m_3}{M_2} \vec{R}|^3} - Gm_2 m_3 \frac{\vec{S} - \frac{m_1}{M_2} \vec{R}}{|\vec{S} - \frac{m_1}{M_2} \vec{R}|^3}$$

$N=4$ Jacobi

$$M_4 = m_1 + m_2 + m_3 + m_4$$

$$\vec{R}_3 = \vec{r}_4 + \vec{R}_3^c = \vec{r}_4 - \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3}{M_3}$$

$N=5$ Jacobi

$$M_5 = M_4 + m_5$$

$$\vec{R}_4 = \vec{r}_5 + \vec{R}_4^c = \vec{r}_5 - \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + m_4 \vec{r}_4}{M_4}$$